Experiment:17 System-Level Simulation and Performance Analysis of a Microwave Radar for Target Detection, Range Estimation and Received-Signal Characterization

Aim: To simulate a microwave radar system and analyze target detection, range estimation, and received-signal characteristics using MATLAB.

Objective:

To simulate a microwave radar system using MATLAB and analyze target detection, target range estimation, and received-signal characteristics through signal processing techniques.

Apparatus:

Computer/Laptop

Matlab Software

Procedure:

- Open MATLAB and create a new script.
- Define the radar parameters, including carrier frequency, bandwidth, pulse duration, and sampling frequency.
- Generate the transmitted chirp signal.
- Calculate the target round-trip delay using the specified target distance.
- Generate the received signal and add noise to simulate practical radar conditions.
- Plot the transmitted and received signals.
- Apply matched filtering to estimate the target range.
- Calculate the radar range resolution and plot the received-signal spectrum.
- Run the program and observe the estimated range and spectral characteristics.

Code:

```
clc;

clear;

close all;

% Radar parameters

c = 3e8;

fc = 10e9;

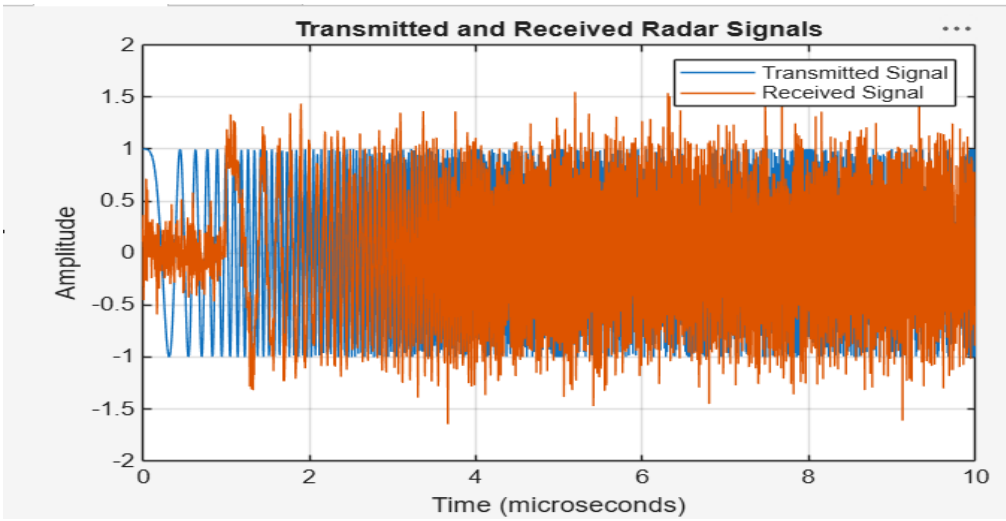
B = 100e6;

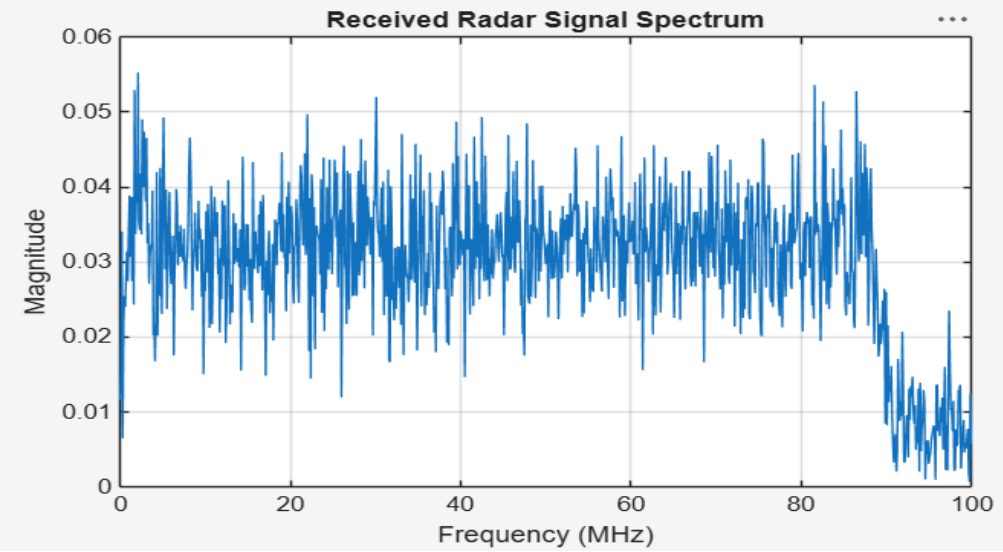
T = 10e-6;
```

```
Fs = 200e6;
R_target = 150;
A = 1;
% Time vector
t = 0:1/Fs:T-1/Fs;
% Transmitted chirp signal
K = B/T;
tx = cos(2*pi*(fc*t + 0.5*K*t.^2));
% Target delay and received signal
tau = 2*R_target/c;
td = t - tau;
rx = zeros(size(t));
idx = td >= 0;
rx(idx) = A*cos(2*pi*(fc*td(idx) + 0.5*K*td(idx).^2));
% Add noise
rx = rx + 0.2*randn(size(rx));
% Plot transmitted and received signals
figure;
plot(t*1e6, tx);
hold on;
plot(t*1e6, rx);
grid on;
xlabel('Time (microseconds)');
ylabel('Amplitude');
title('Transmitted and Received Radar Signals');
legend('Transmitted Signal','Received Signal');
% Range estimation using matched filtering
```

```
mf = conv(rx, fliplr(tx), 'full');
[~, peak] = max(abs(mf));
lag = peak - length(tx);
R_est = lag*c/(2*Fs);
fprintf('Actual Target Range = %.2f m\n', R_target);
fprintf('Estimated Target Range = %.2f m\n', R_est);
% Received signal spectrum
N = length(rx);
Y = fft(rx);
P = abs(Y/N);
P = P(1:floor(N/2)+1);
P(2:end-1) = 2*P(2:end-1);
f = Fs*(0:floor(N/2))/N;
figure;
plot(f/1e6, P);
grid on;
xlabel('Frequency (MHz)');
ylabel('Magnitude');
title('Received Radar Signal Spectrum');
% Range resolution
range_resolution = c/(2*B);
fprintf('Range Resolution = %.2f m\n', range_resolution);
```

Output:





Result:

The microwave radar system is simulated using MATLAB to study target detection, range estimation, and received-signal characteristics.